

BK8520 Electronic Load REST API Flow Diagram

This Mermaid diagram illustrates the POST and GET API endpoints for the BK8520 Electronic Load Control system.

API Architecture Overview

```

graph TB
    subgraph "Client Applications"
        WEB[Web Interface]
        PYTHON[Python Client]
        CURL[cURL/CLI]
        JS[JavaScript Client]
    end

    subgraph "BK8520 REST API Server"
        subgraph "Health & System"
            HEALTH["GET /api/health  
Check service status  
Returns: API health info"]
        end

        subgraph "Device Management"
            CONNECT["POST /api/device/connect  
Connect to BK8520  
Params: port, baudrate  
Action: Establish serial connection"]
            DISCONNECT["POST /api/device/disconnect  
Disconnect device  
Params: none  
Action: Close serial connection"]
            STATUS["GET /api/device/status  
Get connection status  
Returns: connection state, device status"]
            INFO["GET /api/device/info  
i Get device details  
Returns: model, firmware, capabilities"]
            READINGS["GET /api/device/readings  
s Get live measurements  
Returns: voltage, current, power"]
        end

        subgraph "Device Control"
            INPUT["POST /api/device/input  
s Control load input  
Params: enabled (true/false)  
Action: Turn input on/off"]
            CURRENT["POST /api/device/current  
s Set discharge current  
Params: current (amperes)  
Action: Configure CC mode current"]
            VOLTAGE["POST /api/device/voltage  
s Set cutoff voltage  
Params: voltage (volts)  
Action: Set battery protection limit"]
            MODE["POST /api/device/mode  
s Set operation mode  
Params: mode (0=CC,1=CV,2=CW,3=CR)  
Action: Change load operation mode"]
            MAXCURRENT["POST /api/device/max-current  
Set current limit  
Params: current (amperes)  
Action: Configure safety limit"]
            MAXPOWER["POST /api/device/max-power

```

```

    Set power limit
    Params: power (watts)
    Action: Configure safety limit"]
            MAXVOLTAGE["POST /api/device/max-voltage

    Set voltage limit
    Params: voltage (volts)
    Action: Configure safety limit"]
            SETUP["POST /api/device/setup-discharge

❖ Setup CC discharge
    Params: current (amperes)
    Action: Enable constant current mode"]
        end

        subgraph "Battery Testing"
            TESTSTART["POST /api/battery-test/start
        Start capacity test
        Params: discharge_current, cutoff_voltage
        Action: Begin automated test"]
            TESTSTOP["POST /api/battery-test/stop
        Stop active test
        Params: none
        Action: Halt test, save results"]
            TESTSTATUS["GET /api/battery-test/status
        Get test progress
        Returns: test state, measurements
        Updates: capacity, elapsed time"]
            TESTRESULTS["GET /api/battery-test/results
        Get complete results
        Returns: full test data
        Includes: capacity, duration, summary"]
        end
    end

    subgraph "BK8520 Hardware Device"
        DEVICE["BK8520 Electronic Load
        Serial Communication
        Port: /dev/ttyUSB0
        Baudrate: 4800
        Max: 120V, 60A, 999W"]
    end

    %% Client connections
    WEB --> HEALTH
    WEB --> CONNECT
    WEB --> STATUS
    WEB --> READINGS
    WEB --> INPUT
    WEB --> TESTSTART
    WEB --> TESTSTATUS

    PYTHON --> CONNECT
    PYTHON --> CURRENT
    PYTHON --> VOLTAGE
    PYTHON --> TESTSTART
    PYTHON --> TESTRESULTS

    CURL --> HEALTH

```

```

CURL --> INFO
CURL --> MODE

JS --> READINGS
JS --> TESTSTATUS

%% API to Device connections
CONNECT --> DEVICE
STATUS --> DEVICE
INFO --> DEVICE
READINGS --> DEVICE
INPUT --> DEVICE
CURRENT --> DEVICE
VOLTAGE --> DEVICE
MODE --> DEVICE
MAXCURRENT --> DEVICE
MAXPOWER --> DEVICE
MAXVOLTAGE --> DEVICE
SETUP --> DEVICE
TESTSTART --> DEVICE
TESTSTOP --> DEVICE
TESTSTATUS --> DEVICE

%% Styling
classDef getEndpoint fill:#e1f5fe,stroke:#0277bd,stroke-width:2px
classDef postEndpoint fill:#f3e5f5,stroke:#7b1fa2,stroke-width:2px
classDef device fill:#fff3e0,stroke:#f57c00,stroke-width:3px
classDef client fill:#e8f5e8,stroke:#388e3c,stroke-width:2px

class HEALTH,STATUS,INFO,READINGS,TESTSTATUS,TESTRESULTS getEndpoint
class CONNECT,DISCONNECT,INPUT,CURRENT,VOLTAGE,MODE,MAXCURRENT,MAXPOWER,MAXVOLTAGE,SETUP,TESTSTART,TESTSTOP postEndpoint
class DEVICE device
class WEB,PYTHON,CURL,JS client

```

API Endpoint Details

GET Endpoints (Data Retrieval)

Endpoint	Purpose	Response Data
GET /api/health	Service health check	Status and version info
GET /api/device/status	Connection and operational status	Connection state, device status
GET /api/device/info	Device information	Model, firmware, capabilities
GET /api/device/readings	Real-time measurements	Voltage, current, power values
GET /api/battery-test/status	Active test status	Test progress, measurements
GET /api/battery-test/results	Complete test results	Full test data and summary

POST Endpoints (Control Operations)

Endpoint	Purpose	Parameters
POST /api/device/connect	Connect to device	Port, baudrate, timeout
POST /api/device/disconnect	Disconnect device	None
POST /api/device/input	Control input on/off	enabled (boolean)
POST /api/device/current	Set discharge current	current (amperes)
POST /api/device/voltage	Set cutoff voltage	voltage (volts)
POST /api/device/mode	Set operation mode	mode (0=CC, 1=CV, 2=CW, 3=CR)
POST /api/device/max-current	Set current limit	current (amperes)
POST /api/device/max-power	Set power limit	power (watts)
POST /api/device/max-voltage	Set voltage limit	voltage (volts)
POST /api/device/setup-discharge	Setup CC discharge	current (amperes)
POST /api/battery-test/start	Start battery test	discharge_current, cutoff_voltage, max_time_hours
POST /api/battery-test/stop	Stop battery test	None

Data Flow Examples

Device Connection Flow

```
sequenceDiagram
    participant Client
    participant API
    participant Device

    Client->>API: POST /api/device/connect
    Note right of Client: {"port": "/dev/ttyUSB0", "baudrate": 4800}

    API->>Device: Serial connection attempt
    Device-->>API: Connection confirmation
    API->>Device: Setup basic test parameters
    Device-->>API: Setup confirmation

    API-->>Client: {"success": true, "connected": true}

    Client->>API: GET /api/device/info
    API->>Device: Request device information
    Device-->>API: Model, firmware, capabilities
    API-->>Client: Device info response
```

Battery Test Flow

```
sequenceDiagram
    participant Client
    participant API
    participant Device

    Client->>API: POST /api/battery-test/start
    Note right of Client: {"discharge_current": 2.0, "cutoff_voltage": 10.5}

    API->>Device: Set current and voltage parameters
    Device-->>API: Parameters set
    API->>Device: Enable input (start discharge)
    Device-->>API: Input enabled

    API-->>Client: {"test_active": true}

    loop Monitoring
        Client->>API: GET /api/battery-test/status
        API->>Device: Read current measurements
        Device-->>API: Voltage, current, power
        API-->>Client: Test progress and measurements
    end

    Note over API,Device: Test stops when cutoff voltage reached
    API->>Device: Disable input
    API-->>Client: Test completed with results
```


System Architecture

The API server acts as a bridge between client applications and the physical BK8520 device:

- **FastAPI Server:** Handles HTTP requests and provides OpenAPI documentation
- **Serial Communication:** Direct communication with BK8520 via RS232/USB
- **Real-time Monitoring:** Continuous measurement reading during tests
- **Data Persistence:** Test results saved to JSON files
- **Error Handling:** Comprehensive error reporting and device safety checks

Base URL: <http://10.100.10.190:8000>